

DAY 6 INTERNSHIP REPORT — EXCEL FORMULAS & FUNCTIONS

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Executive Overview

On Day 6 of the internship, the focus shifted to mastering foundational and advanced Excel formulas using the Sample Superstore dataset. A comprehensive, multi-sheet workbook was developed to demonstrate practical business applications of different function families, moving from raw data to a fully interactive and error-proofed analysis.

Logical & Lookup Functions

To automate decision-making and data retrieval, several logical and referencing formulas were applied:

- Implemented IF and IFS functions to categorize orders by profitability and margin tiers, such as High, Medium, or Loss.
- Built custom logical tests to automatically flag heavily discounted orders for manual review.
- Wrapped failure-prone calculations in IFERROR to ensure clean outputs instead of standard Excel error codes.
- Applied VLOOKUP and XLOOKUP for both exact matches and approximate band-matching against a discount reference table.
- Executed a two-way INDEX/MATCH to extract precise profit figures from a dynamic Region by Category matrix.

Dynamic Aggregation

To enable high-level business intelligence without relying solely on pivot tables, a custom aggregation dashboard was constructed:

- Built an interactive summary panel driven by data validation dropdown inputs for Region and Category.
- Leveraged SUMIFS to calculate conditional financial metrics, such as total loss dollars at discounts greater than 40%.
- Utilized COUNTIFS to isolate the exact number of loss-making 'Tables' orders across all regions.
- Applied AVERAGEIFS to determine the average discount specifically on loss-making Consumer orders.

Data Cleaning & Text Manipulation

To ensure data integrity prior to analysis, text functions and edge-case rules were established:

- Concatenated geographic fields into clean, unified strings using the TEXTJOIN function.
- Built an automated ZIP code validation check using the LEN function to flag entries with missing digits.
- Extracted specific string segments using LEFT and RIGHT functions to parse text arrays.
- Identified and documented critical edge cases that break standard formulas, including trailing spaces, blanks, and numbers stored as text.